

Exercise - LLMs

MIR IT-Protection WS25

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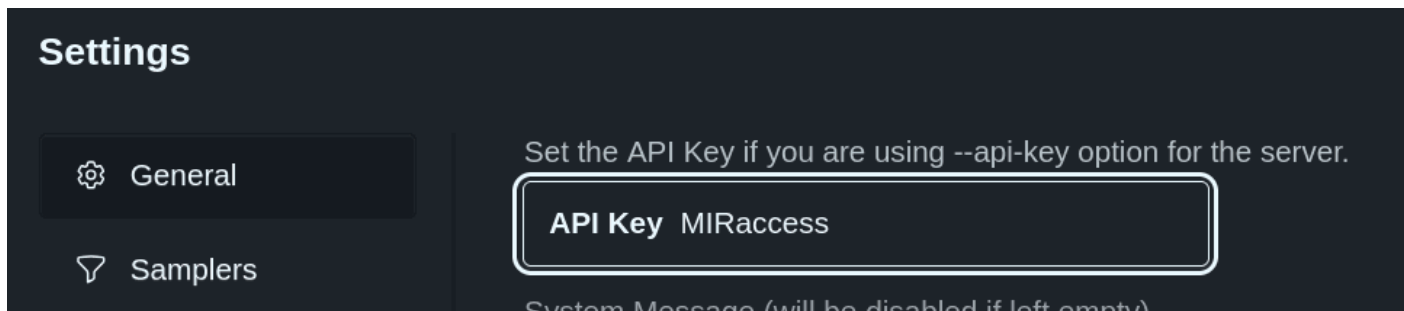
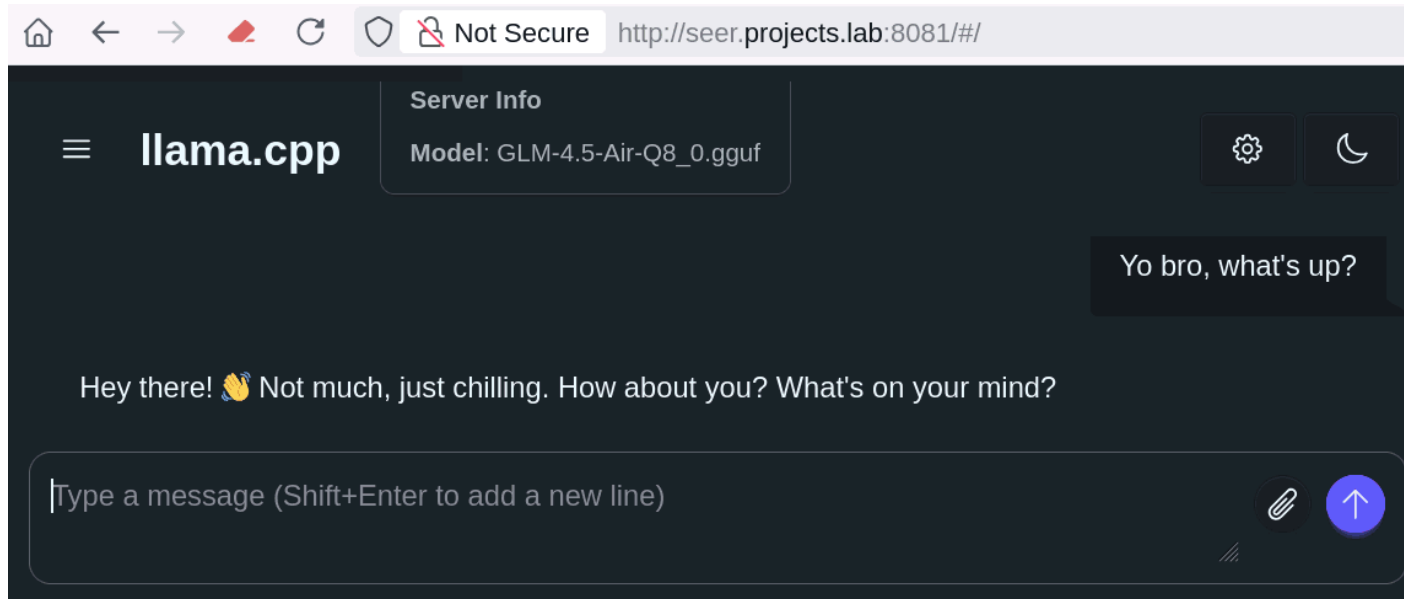
- Investigate the capabilities and limitations of different LLMs
- There are locally running LLMs in our lab available for testing – with different features, capabilities, quantizations, speed, ...
- This provides a controlled, repeatable environment – not like commercial providers (e.g. ChatGPT) that tweak their models silently... (bad for controlled experiments for research ☹)

Available LLMs / Models

Context	Port	ApiKey	Model
32768tk	8081	MIRaccess	GLM-4.5-Air-Q8_0.gguf
16384tk	8084	MIRaccess	gpt-oss-120b-F16.gguf
16384tk	8085	MIRaccess	gemma-3-27b-it-Q8_0.gguf
8192tk	8091	MIRaccess	gemma-3-1b-it-Q8_0.gguf
8192tk	8092	MIRaccess	gemma-3-4b-it-Q8_0.gguf
8192tk	8093	MIRaccess	gemma-3-12b-it-Q8_0.gguf

Note: Access requires either being on a lab's PC, or accessing them remote via VPN (https://start.nwt.fhstp.ac.at/wiki/doku.php/nwtlab:openvpn_en)

Use via Webbrowser / WebUI



Use Cmdline Client in Lab

```
mpirker@pc25 ~$ source /netexport/public/lecturers/mpirker/latest/environment
FORGE mpirker@pc25 ~$ askllm --help
chat      run one chat
info      query model information at server
version   build information
FORGE mpirker@pc25 ~$ askllm info --server http://seer.projects.lab:8084
60.87GB gpt-oss-120b-F16.gguf http://seer.projects.lab:8084
FORGE mpirker@pc25 ~$ askllm chat --help
-s, --server <URL>          [default: http://localhost:8080]
-k, --api-key <API_KEY>     API key for authentication (or from LLAMA_API_KEY)
--sp <prompt string>       the system prompt
--up <prompt string>       the user prompt
--spf <filename>           file to read system prompt from
--upf <filename>           file to read user prompt from
--nothinkglm               disable thinking for GLM4.5
--raw                      disable any output post processing
--temp <x.x>               set model randomness of the generated text [default: 0.0]
--max-req <chars>          limit request length to max chars [default: 0]
--grammar <filename>       file to read grammar constraints from
--seed <integer>           seed for reproducible outputs (-1 for random) [default: -1]
--timeout <seconds>        HTTP request timeout in seconds [default: 30]
--top-p <x.x>              set model top-p sampling [default: 0.9]
-h, --help                 Print help
FORGE mpirker@pc25 ~$ askllm chat --server http://seer.projects.lab:8084 --api-key MIRaccess
--sp "You are a wise sage that answers in short, cryptic sentences" --up "What is the meaning
of life?"
In the echo of breath, purpose whispers.
FORGE mpirker@pc25 ~$
```

- Run experiments **with these local LLMs** and document your insights.
 - Successes, of course, but also what you tried and it didn't work out.
- Check also the different settings and how they affect randomization/quality of your answers (e.g. temperature). Try changing them on your own.
- Note that GLM is a “thinking” model by default and takes a longer time to answer (which may result in a connection timeout).
- These LLMs were trained with data that was available some time ago, they cannot know e.g. about recent news events.

- Find an example where the answer quality noticeably changes/degrades/becomes wrong/unuseable/...
...when going through different Gemma models (27b→12b→4b→1b)
- Can you find an example that shows where each model (GLM, GPT-OSS and Gemma) is noticeably good, and where it is not competent?
- Can you find examples where LLMs blatantly lie to you?
- Can you find examples where LLMs are surprisingly “super-human” in what they can do? A real help?

- Can you find examples where LLMs surprisingly fail at a seemingly trivial task (for humans)?
- By using LLMs for research-like questions, are the references LLMs provide real sources or are citations made up? Positive/negative example?
- Ask the LLMs something you have good knowledge of (e.g. your Bachelor theses topic) – are the LLMs providing competent answers? (don't try asking things that are easily available from Wikipedia)

- Think of additional, interesting experiments of your own...

- Individual work
- Document your findings in a report
 - ~5 pages
 - no specific style required (but use normal font size, margins, etc.)
- Upload as PDF (filename Familyname-LLMs.pdf) on ECampus
- ...until **Wed Oct 15th 23:59**
(the LLMs will have their password changed after that – others want to use them, too...)

Thank you! Questions?

Feedback welcome
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